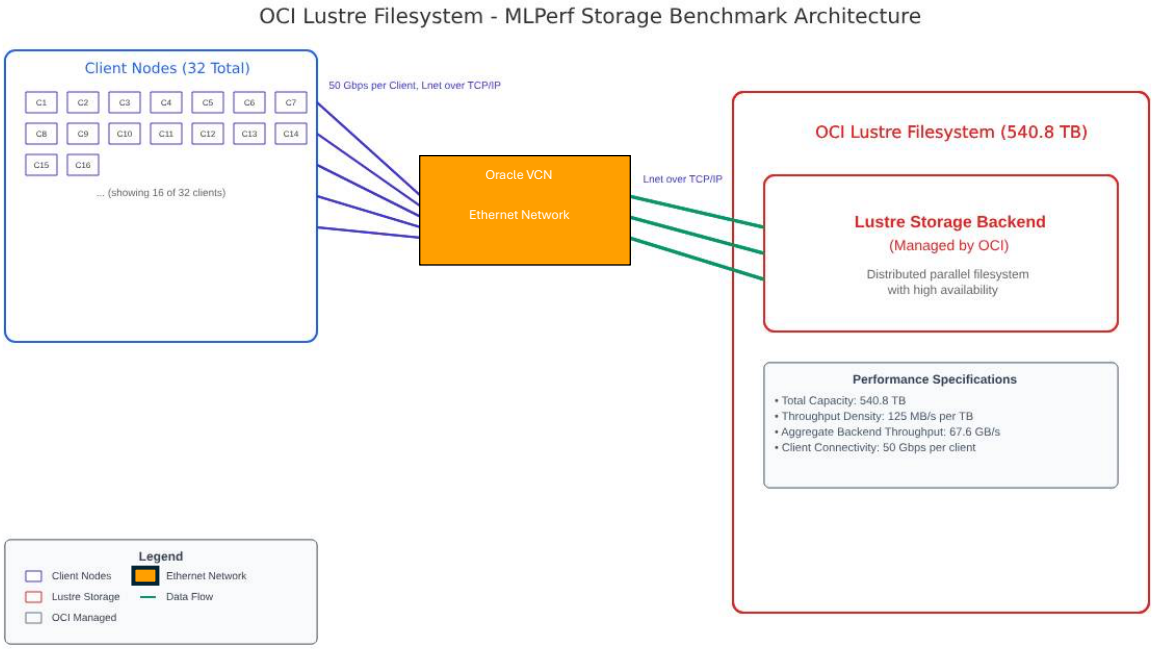
Oracle Cloud Infrastructure (OCI) Lustre Filesystem MLPerf Storage Benchmark System Configuration



1. Overview of OCI Managed Lustre File System (LFS)

Oracle Cloud Infrastructure (**OCI**) provides a fully managed Lustre File System (LFS) that is designed for high availability and high performance. OCI LFS enables customers to scale capacity and performance according to their workload requirements.

Key features include:

- Managed service with minimal administrative overhead.
- High availability, fault tolerance, and high scalability.
- Flexible capacity and performance provisioning, with performance tiers delivered in MB/s per TB (125, 250, 500, and 1000 MB/s/TB).
- Integration with other OCI services including networking services (VCN), OCI metrics, Identity and Access Management, and compute to streamline deployment and management.
- Deployments using OCI console, CLI, API, or Terraform.

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2. Cluster Setup and Benchmark Execution

The MLPerf Storage benchmarks were executed on an OCI cluster setup as illustrated in the provided diagram. The architecture consisted of multiple compute/client nodes connected via a high-speed 50 Gbps Ethernet interconnect to the managed Lustre filesystem backend.

Benchmarks were executed from the client nodes against the shared Lustre file system. Cluster comprised of a total of 32 client nodes. Each MLPerf benchmark was run with a different number of clients and accelerator combinations based on the workload characteristics in the same cluster.

3. Benchmark Configuration: OCI LFS Storage Details:

- **Performance Tier (MB/s/TB):** 125
- **Provisioned Capacity (TB):** 540.8
- **Aggregate throughput (MB/s):** 67,600

4. Benchmark Configuration: Client Nodes

All client nodes used for the benchmarks were OCI **BM.Optimized3.36** instances. Lustre client version 2.15.5 was installed on the clients via the default OCI Lustre client installation.

System Details:

- **CPU Architecture**: `Intel(R) Xeon(R) Gold 6354 CPU @ 3.00GHz`
- **Memory per Node**:

a. For Training Benchmarks :

96GB (Original: 512GB, Reduced to 96GB for all Training benchmarks via adding mem=96G to the grub file. (`GRUB_CMDLINE_LINUX_DEFAULT="quiet splash mem=96G"`)

b. For Checkpointing Benchmarks:

512GB

- **Number of Cores per Node**: 36
- **Number of Sockets**: 2
- **Cores per socket**: 2
- **Threads per core**: 2

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- **CPU Max MHz**: 3600
- **CPU Min MHz**: 800
- **CPU(s)**: 36 (with SMT enabled, 72)
- **Operating System**: `Ubuntu 22.04.5 LTS`
- **Kernel**: `5.15.0-1074-oracle`
- **Lustre Client Version**: `2.15.5-oci0-12-gd3b2033-1`
- **Client Interconnect Speed**: 50Gbps, MTU 9000, Mellanox Technologies MT2892 Family [ConnectX-6 Dx]
- **LNet Configuration**: `tcp(ens300f0np0),` TCP/IP ethernet network.

Results submitted:

A total of 5 training benchmarks were submitted as outlined below.

Benchmark	Memory on Clients	# Clients	# Accelerators/Node	Total Accelerators	Mode
ResNet50	96	24	14	336	H100
Unet3d	96	11	4	44	A100
Unet3d	96	11	2	22	H100
Cosmoflow	96	20	6	120	A100
CosmoFlow	96	25	3	75	H100

A total of 2 Checkpointing benchmarks were submitted as outlined below:

Benchmark	Memory on Clients	# Clients	Total Processes
Llama-8b	512GB	1	8
Llama-405b	512GB	32	512

5. Tunings and Settings Applied for Training Benchmarks

For benchmarks run against 96GB (client/compute node) memory, following tunings were applied for all benchmarks:

MDT Striping and Tuning:

- `lfs setdirstripe -c -1 -D /mnt/lustrefs/${benchmark_directory}`
- `lctl set_param mdc.*.max_rpcs_in_flight=32`
- `lctl set_param *.*.*MDT*.lru_max_age=1000000`
- `lctl set_param *.*.*MDT*.lru_size=11000`

LNet Configuration and Tunings:

- `options ksocklnd credits=2048 peer_buffer_credits=1024 tx_buffer_size=65536 rx_buffer_size=65536 keepalive=10 conns_per_peer=16 peer_credits=16`

For Cosmoflow benchmark:

Lustre Client Tunings:

- `lctl set_param ldlm.namespaces.*.lru_max_age=10000`

For Unet3d benchmark:

- `lfs setstripe -S 20M /mnt/lustrefs/${benchmark_directory}`

No additional system tuning was applied for the training benchmarks. No code modifications or changes were made. All results presented correspond to closed submissions.

5. Tunings and Settings Applied for Checkpointing Benchmarks

For all Checkpointing benchmarks:

Sysctl Tunings:

- `sudo sysctl -w vm.dirty_ratio=10`
- `sudo sysctl -w vm.dirty_background_ratio=4`
- `sudo sysctl -w vm.dirty_expire_centisecs=400`
- `sudo sysctl -w vm.dirty_writeback_centisecs=100`

Lnet Config:

- `options ksocklnd credits=2048 peer_buffer_credits=1024 tx_buffer_size=65536 rx_buffer_size=65536 keepalive=10 conns_per_peer=16 peer_credits=16`

For llama3-8b:

- `sudo lfs setstripe -S 20M /mnt/lustrefs/${benchmark_directory}`

For llama3-405 benchmark:

Lustre Tunings:

- `sudo lctl set_param llite.*.max_cached_mb=8192`
- `sudo lctl set_param osc.*.max_dirty_mb=4`
- `sudo lctl set_param llite.*.max_read_ahead_mb=64`
- `sudo lctl set_param llite.*.max_read_ahead_per_file_mb=64`
- `sudo lctl set_param llite.*.max_read_ahead_whole_mb=4`
-

No additional system tuning was applied for the Checkpointing benchmarks. No code modifications or changes were made. All results presented correspond to closed submissions.

Submission Details

System Name of the submission: Oracle-540.8TB-OCI-LFS-125MBperSec-perTB-1x50GbpsEthernet-Clients

Submitters Organization: Oracle

Submission Date: July 10th, 2025

Version of the Benchmark: mlpstorage v2.0

Submission Division: Closed

Solutions Type of the submission: OCI LFS (OCI Lustre Filesystem)

System Architecture: Cloud

Usable Capacity: 540.8 TB